

Legends

Preparatory Sessions

Primers

EPAT Lectures

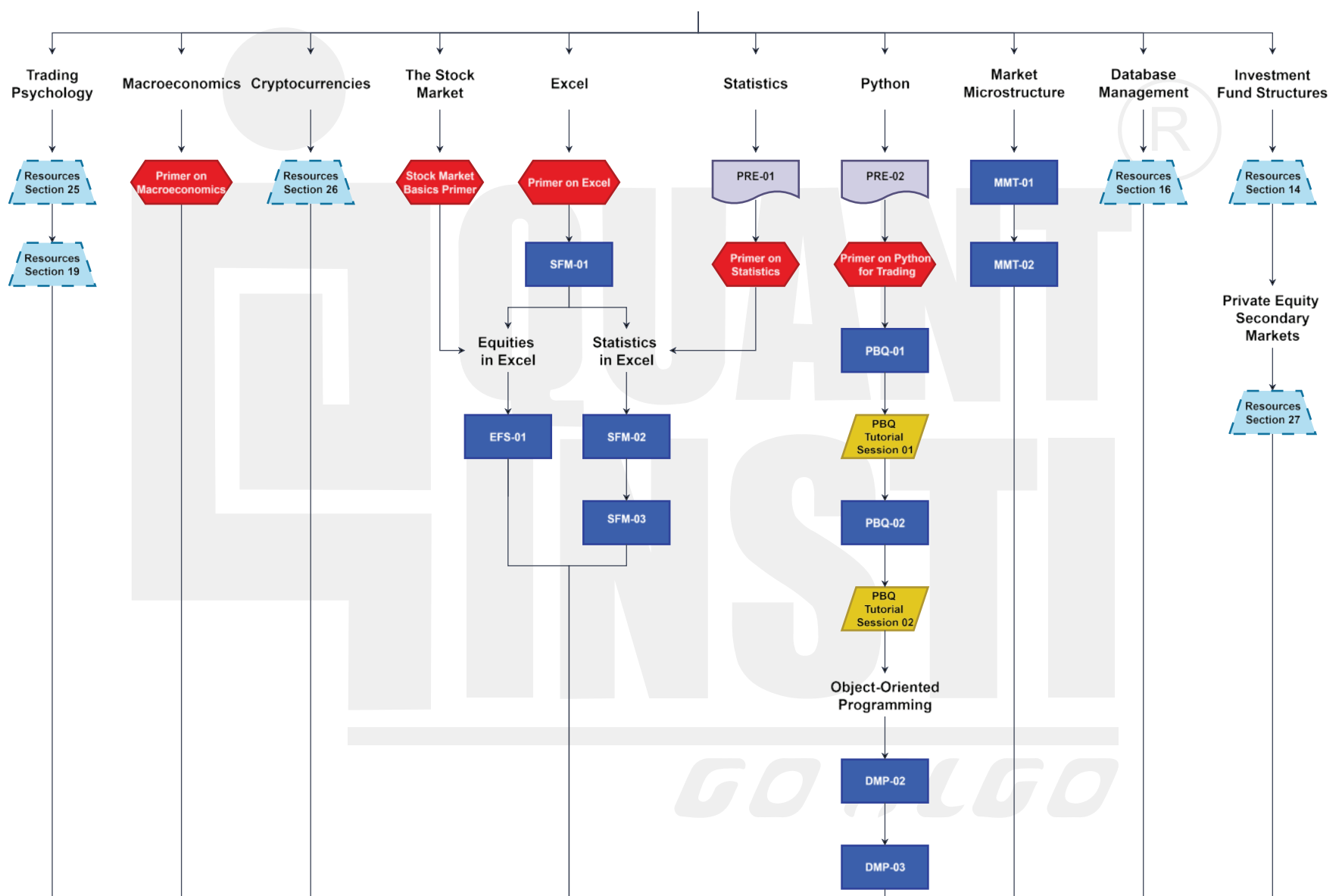
Optional Sessions

EPAT Resources

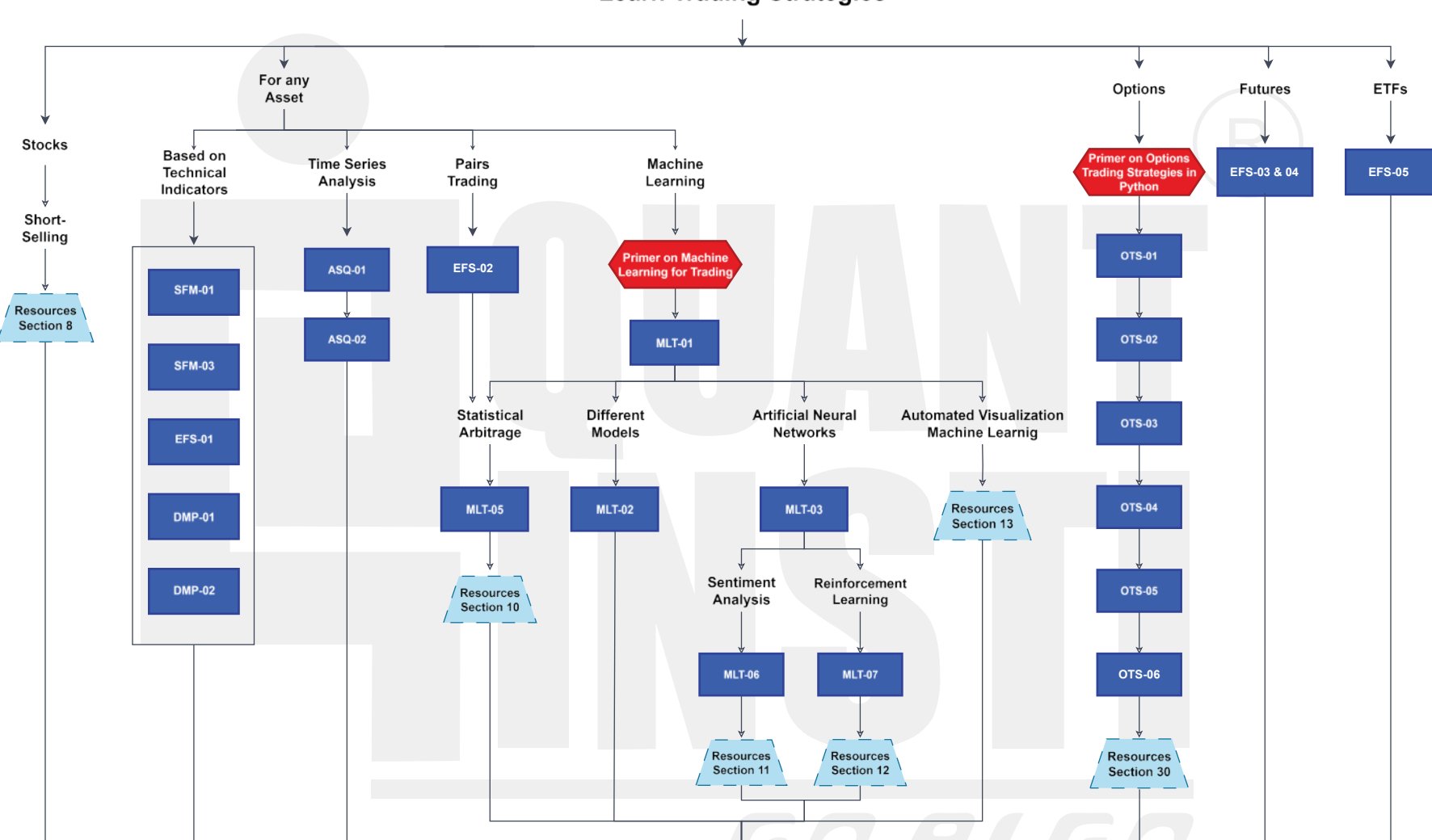
Mini Projects

Go Algo!

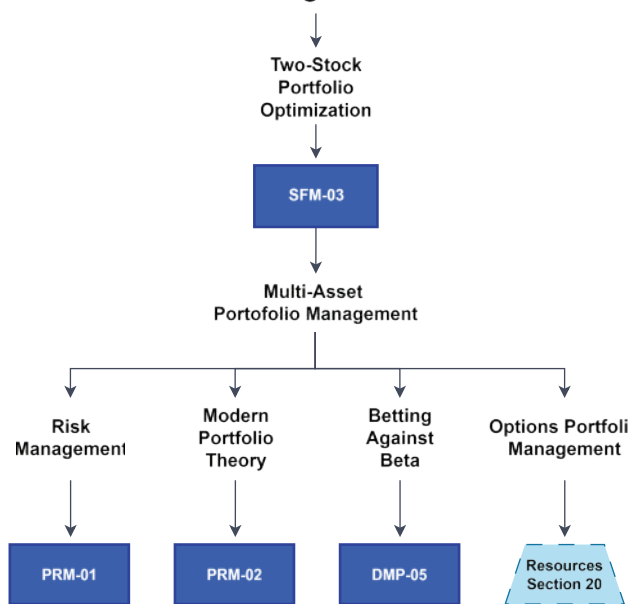
Laying the Foundations



Learn Trading Strategies



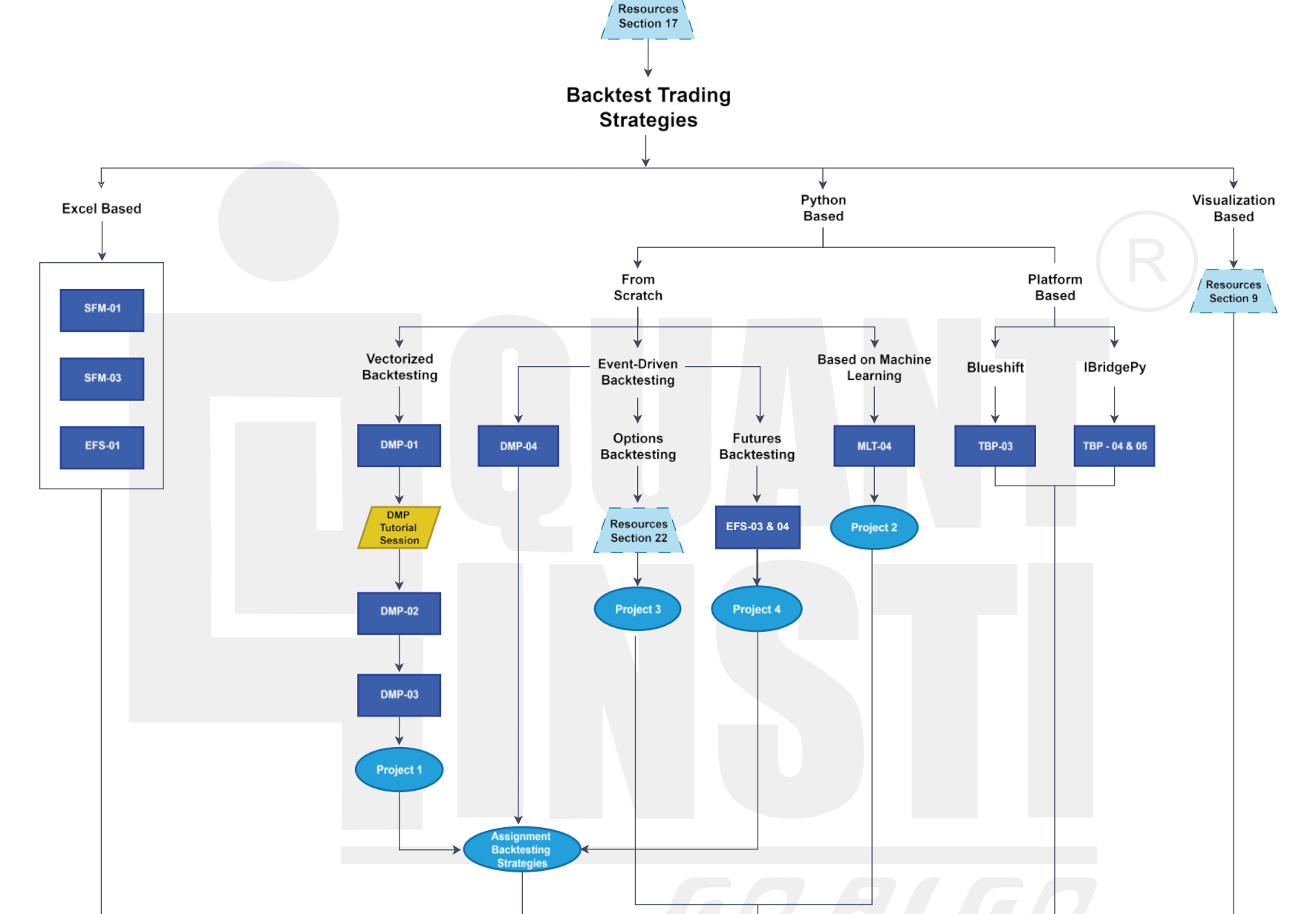
Portfolio Construction and Risk Management



Practical Quant Insights

Resources Section 17

Backtest Trading Strategies



Create a Trade Sheet

Resources Section 29

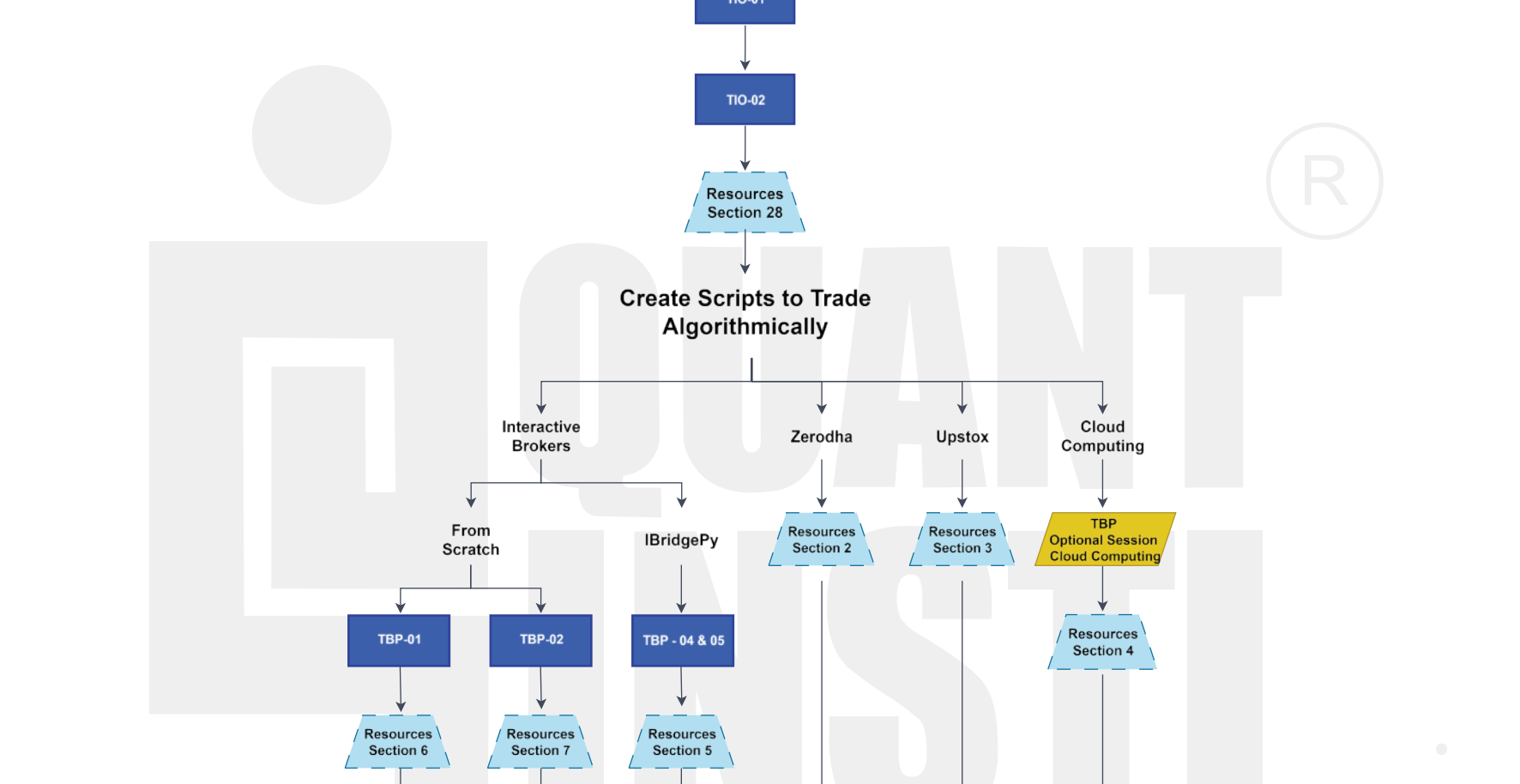
Set Up a Trading Desk

TIO-01

TIO-02

Resources Section 28

Create Scripts to Trade Algorithmically



Execution Strategies

MMT-03

MMT-04

Go Live!

From Scratch

TBP Live Trading Using IB API

IBridgePy

TBP - 04 & 05

Performance Evaluation

For any Asset

EFS-01

For Options

OTS-05

Happy Trading!

Get a Quant Job

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The EPAT Curriculum: A Guide to Content Interlinking

This guide maps out your learning path in the Executive Programme in Algorithmic Trading (EPAT), structured around the 12 key phases of your EPAT course map. It explains how each module and lecture connects to the next, giving you a high-level view of your journey.

Phase 1: Laying the Foundations

This initial phase is designed to build your core skill set. Here, you'll cover the quantitative, programming, and market knowledge that every subsequent phase will rely upon.

Contributing Modules & Lectures

- **Statistics for Financial Markets (SFM)**
 - **Module Overview:** In SFM, you'll build your foundation in algorithmic trading. You will start by using spreadsheets to learn the statistical principles that underpin trading strategies, helping you move from intuition to a data-driven understanding of market behavior.
 - **Intra-Module Lecture Progression:**
 - **SFM-01 to SFM-02:** Now, after you learn the "how" of basic calculations in Excel in **SFM-01**, **SFM-02** provides the "why" by introducing the core statistical theory behind those numbers. This ensures your practical skills are backed by a solid conceptual understanding.
 - **SFM-02 to SFM-03:** Then, as a final lecture here, **SFM-03** takes the formal statistical theory you learned in **SFM-02** and applies it back into the practical Excel environment. This allows you to perform more advanced financial analysis and solidifies your understanding by bridging theory and practice.
- **Market Microstructure for Trading (MMT)**
 - **Inter-Module Linking:** Next, regarding the vast topic of market microstructure, this module builds directly on the statistical foundations of SFM; so, do not skip the previous module! While SFM teaches you how to analyze market behavior, MMT explains the mechanics of how that behavior translates into the process of actual trades. Understanding concepts like bids, offers, and liquidity is necessary to appreciate the real-world costs that affect the statistical outcomes you've learned to measure.
 - **Intra-Module Lecture Progression:**

- **MMT-01 (Intro to Execution Strategies)** provides you with the fundamental vocabulary of the market, such as orderbooks and basic order types. This knowledge is a prerequisite for understanding the more complex execution logic that follows. *Side note: In Phase 9, you'll see a more detailed breakdown of how the advanced MMT lectures build upon this foundation. This part is a lecture progression within the module.*
- **Python: Basics & Its Quant Ecosystem (PBQ)**
 - **Inter-Module Linking:** This module introduces Python as a useful tool for implementing the knowledge from the first two modules. It will give you a decent understanding of programming. It will enable you to understand how to program everything you'll need for your strategy ideation later.
 - **Intra-Module Lecture Progression:**
 - **PBQ-01 to PBQ-02:** After learning the general syntax and structure of Python in **PBQ-01**, **PBQ-02** focuses your skills on the specific tools and libraries, like Pandas, that are used for financial analysis. This transition helps you move from a general programmer to a quantitative analyst.
- **Data Analysis & Modeling in Python (DMP)**
 - **Inter-Module Linking:** This module is the next step. It's a direct application of the skills you acquired previously. You'll use your Python skills from PBQ to implement backtesting frameworks. Your existing statistical knowledge from SFM will be used to calculate returns and evaluate performance. Your understanding of market mechanics from MMT will inform the creation of more realistic backtesters.
 - **Intra-Module Lecture Progression:**
 - **DMP-01 to DMP-02:** **DMP-02** elevates the simple, procedural backtesting script you wrote in **DMP-01** into a more robust and scalable Object-Oriented Programming (OOP) framework.

QUANT INSIGHT: The Three Pillars

The journey into quantitative trading begins with three essential pillars: a solid grasp of statistics to analyze data, proficiency in Python to automate tasks, and an understanding of market mechanics to know the rules of the game. Mastering these foundations is the prerequisite for any meaningful strategy development.

Phase 2: Learn Trading Strategies

This is the most extensive phase, where you'll apply your foundational knowledge to ideate, build, and analyze a wide variety of trading strategies.

Contributing Modules & Lectures

- **Equity, FX & Futures Strategies (EFS)**

- **Inter-Module Linking:** This module integrates your prior lectures from above to focus on creating specific trading strategies. You'll rely on your Excel and Python skills from SFM and DMP, respectively, to build and backtest ideas, your statistical knowledge from SFM to develop quantitative strategies, and your understanding of market microstructure from MMT to ensure your strategies are in line with existing market realities.
- **Intra-Module Lecture Progression:**
 - **EFS-01 & 02 to EFS-03:** After learning the universal components of a trading system in **EFS-01 & 02**, **EFS-03** allows you to apply that framework to a specific, statistically-intensive strategy, moving from general theory to a concrete example.
 - **EFS-03 to EFS-04 & 05:** **EFS-04 & 05** introduce momentum strategies, which are fundamentally different from the mean-reversion focus of **EFS-03**. This contrast is designed to expand your strategic toolkit and help you recognize different types of market behavior.

- **Advanced Statistics for Quant Strategies (ASQ)**

- **Inter-Module Linking:** You'll use your Python skills from PBQ and DMP to implement time series models (like ARIMA and GARCH) that enhance the strategies developed in EFS. It allows for more adaptive responses to changing market conditions.
- **Intra-Module Lecture Progression:**
 - **ASQ-01 to ASQ-02:** **ASQ-01** teaches you to build models that predict market direction. **ASQ-02** then enhances these models by adding volatility forecasting, allowing you to create more complete systems that adapt to both price movement and market risk. ARIMA to predict stock returns and GARCH to predict stock volatility.

- **Machine Learning for Trading (MLT)**

- **Inter-Module Linking:** This module leverages your entire preceding curriculum. You'll use your statistical knowledge from SFM and ASQ as a basis for understanding ML algorithms, and your Python skills from PBQ and DMP to

implement models and engineer predictive features.

AHA! MOMENT: From Statistics to Machine Learning

The key realization here is that Machine Learning is not magic; it's a powerful extension of statistics. You'll see that ML models are fundamentally sophisticated pattern-recognition systems that build upon the same principles of probability and regression you learned earlier, but they are capable of finding more complex, non-linear relationships in the data.

- **Intra-Module Lecture Progression:**

- **MLT-01, 02, 03 to MLT-04:** After learning the individual ML algorithms in the first three lectures, **MLT-04** is an application step that shows you how to integrate them into a complete, end-to-end strategy development workflow.
- **MLT-04 to MLT-05:** **MLT-05** builds on the workflow from **MLT-04** and the basics of pairs trading in EFS-03 by demonstrating an application of unsupervised learning (PCA) to a known strategy type (statistical arbitrage), showing how ML can enhance traditional quant methods.
- **MLT-05 to MLT-06:** **MLT-06** expands your capabilities beyond the numerical data used in previous lectures by teaching you how to extract trading signals from unstructured text.
- **MLT-06 to MLT-07:** **MLT-07** is the synthesis of the entire module, where you'll use features from all previous lectures (including sentiment from **MLT-06**) to build a fully adaptive Reinforcement Learning (RL) agent.

- **Options Trading & Strategies (OTS)**

- **Inter-Module Linking:** This is your initiation into the world of options trading. Here, you require the statistical foundations from SFM and forecasting models like GARCH from ASQ. You'll use your Python skills from PBQ and DMP to create basic pricing models, build backtesters, and quantify risk.
- **Intra-Module Lecture Progression:**
 - **OTS-01 to OTS-02:** **OTS-01** establishes the general theory of "edge" and "risk." Next, **OTS-02** makes this theory concrete by identifying a specific, quantifiable "edge" in the options market—the volatility premium.
 - **OTS-02 to OTS-03:** After identifying the volatility premium in **OTS-02**, **OTS-03** then provides you with the quantitative models (Black-Scholes, the Greeks) you need to actually measure and price that edge.
 - **OTS-03 to OTS-04:** **OTS-04** moves from analysis to synthesis. It shows you how to use the pricing models from **OTS-03** to structure an actual

trade designed to capture the volatility premium.

- **OTS-01-to-04 to OTS-05:** **OTS-05** explains more about the evaluation of Option trades, how to hedge, and expiration trading.
- **OTS-01-to-05 to OTS-06:** By using all the knowledge from the previous lectures, **OTS-06** provides a framework and examples to backtest an options strategy, with a strategy implementation, performance metrics and analytics, and Risk Management and Optimization.

💡 **QUANT INSIGHT: Building Your Strategic Toolkit**

This phase is where theory is put into practice. The core activity of a quant is to develop and test a diverse set of strategies, from classic statistical arbitrage and momentum to advanced machine learning and options models. A broad toolkit of strategies is essential for adapting to different market conditions.

Phase 3: Portfolio Construction & Risk Management

This phase takes your focus from individual strategies to managing a collection of assets within a robust risk framework.

Contributing Modules & Lectures

- **Portfolio Optimization & Risk Management (PRM)**
 - **Inter-Module Linking:** This module provides a basic framework for managing a collection of strategies or assets. You can apply your individual strategies, developed in EFS, ASQ, MLT, and OTS, and evaluate them within a broader portfolio context. You'll apply statistical tools from SFM and ASQ to measure portfolio-level risk and calculate risk-adjusted returns.

🧠 **AHA MOMENT: The Power of Uncorrelated Returns**

The goal of portfolio management isn't just to find many profitable strategies, but to find profitable strategies that are *uncorrelated* with each other. You'll realize that a portfolio of several moderately profitable, low-correlated strategies is often far more robust and desirable than a portfolio with one highly profitable strategy, because it smoothens out your returns and protects you from ruin when that one strategy inevitably fails.

- **Intra-Module Lecture Progression:**

- **PRM-01 to PRM-02:** **PRM-01** establishes the qualitative framework for risk management, teaching you to identify different types of risk. **PRM-02** builds directly on the previous topic by providing the quantitative tools to actually measure, analyze, and optimize that risk within a portfolio with the Modern Portfolio Theory (MPT).
- **Options Trading & Strategies (OTS)**
 - The lectures in this module are explained in detail in the '**Learn Trading Strategies**' phase. Here, the focus is on applying the risk management aspect of **OTS-05 (Hedging and Trade Evaluation)**, where you learn how to neutralize options risks in a practical setting.

💡 **QUANT INSIGHT: From Strategy Risk to Portfolio Risk**

Be sure that a profitable strategy can still fail if it's part of a poorly constructed portfolio. This phase shifts the focus from individual strategy performance to holistic risk management, teaching you to measure and control the combined risk of all your positions to ensure long-term survival and success.

Phase 4: Practical Quant Insights

We provide here a Resources recording video, but you can also learn some valuable information here.

Examples from Lectures

- In **DMP-02**, you'll learn *why* Object-Oriented Programming is useful for building scalable backtesting frameworks.
- The contrast between **EFS-03** and **EFS-04** gives you insight into the fundamental dichotomy of mean-reversion vs. momentum.
- **ASQ-02** shows you the practical application of combining price-direction and volatility models.
- In **OTS-02**, you'll learn about the persistent volatility premium as a real-world "edge" to be exploited.

💡 **QUANT INSIGHT: Bridging Theory and Practice**

The curriculum is rich in crucial insights that bridge academic theory with real-world application. Understanding concepts like the mean-reversion vs. momentum dichotomy or the persistent

volatility premium is what will enable you to create your own customized edge.

Phase 5: Backtest Trading Strategies

This phase focuses on validating your trading ideas using historical data.

Contributing Modules & Lectures

- **Data Analysis & Modeling in Python (DMP)**

- **Intra-Module Lecture Progression:**

- **DMP-02 to DMP-03:** **DMP-03** takes the OOP framework from **DMP-02** and shows you how to make it fast and efficient by using vectorized operations.
 - **DMP-03 to DMP-04:** **DMP-04** contrasts the vectorized approach from **DMP-03** by introducing a more realistic, tick-by-tick event-driven simulation, which better mimics a live trading environment.

 AHA MOMENT: The Backtester's Dilemma: Speed vs. Realism

You'll discover that there's a fundamental trade-off in backtesting. A vectorized backtester is incredibly fast, allowing you to test thousands of ideas quickly. However, an event-driven backtester is much more realistic, as it helps you avoid look-ahead bias and simulates how a strategy would trade in real-time. Knowing when to use the fast-but-imperfect tool for initial research and when to use the slow-but-accurate tool for final validation is a key skill of a professional quant.

- **DMP-04 to DMP-05:** **DMP-05** takes the backtesting engines you built in the previous lectures and integrates them into a complete quant strategy workflow, from idea generation to final performance analysis.

- **Options Trading & Strategies (OTS)**

- The lectures in this module are explained in detail in the '**Learn Trading Strategies**' phase. Here, the focus is on the practical application of **OTS-06 (Creating an Options Backtester)**, where you learn to build a specialized backtester to systematically test the ideas you developed in the earlier OTS lectures.

- **Trading & Backtesting Platforms (TBP)**

- The lectures in this module are explained in detail in the '**Phase 8: Create Scripts to Trade Algorithmically**' phase. Here, you'll focus on the backtesting capabilities of **TBP-03**, which teaches the use of dedicated platforms

(Blueshift), and **TBP-04 & 05**, which allow for backtesting against actual Interactive Brokers data using IBridgePy.

 QUANT INSIGHT: The Discipline of Validation

An idea is not a strategy until it has been rigorously backtested. This phase is critical for developing the discipline to honestly evaluate your ideas against historical data, using robust frameworks to avoid common biases and ensure your strategy's edge is real, not imagined. You'll learn from experience that even backtesting is not enough for live trading, but it's a necessary condition!

Phases 6 & 7: Create a Trade Sheet & Set up a Trading Desk

This phase covers the practical, operational, and technical aspects of establishing a functional trading desk.

Contributing Modules & Lectures

- **Trading, Tech and Infra & Operations (TIO)**
 - **Inter-Module Linking:** A complex but very useful topic! This module shifts your focus from strategy to trading infrastructure. The risk management principles from PRM are paramount, as robust system architecture is a primary defense against operational risks.
 - **Intra-Module Lecture Progression:**
 - **TIO-01 to TIO-02:** Here, **TIO-01** covers the technical blueprint of a trading system (hardware and software). **TIO-02** builds on the previous lecture by covering the business, compliance, and operational management of that system, which includes creating trade sheets and other necessary documentation.

 QUANT INSIGHT: Trading as a Business Operation

Algorithmic trading is not just about code; it's a business operation. This phase covers the critical infrastructure—from system architecture and hardware to compliance and operational procedures—that forms the professional backbone of any successful trading desk.

Phase 8: Create Scripts to Trade Algorithmically

This phase is about writing the actual code that connects to brokerage systems to automate your trading.

Contributing Modules & Lectures

- **Trading & Backtesting Platforms (TBP)**
 - **Inter-Module Linking:** TBP is one of the last modules of this entire curriculum. You'll directly use your Python skills from PBQ and DMP to connect with brokerage systems. The risk management frameworks from PRM and infrastructure knowledge from TIO are needed for managing a live system.
 - **Intra-Module Lecture Progression:**
 - **TBP-01 to TBP-02:** **TBP-01** teaches you how to connect to a specific platform, the Interactive Brokers API. **TBP-02** then builds on this by broadening your skills to a universal API standard (REST), making your knowledge general enough to many other brokers' APIs.
 - **TBP-02 to TBP-04 & 05:** After learning the low-level API connections, **TBP-04 & 05** introduce you to a higher-level tool (IBridgePy) that simplifies the process of scripting and deploying your algorithms.

QUANT INSIGHT: Connecting Your Code to the Market

This phase provides the final, practical link, teaching you how to use APIs to translate your proven-edge strategy code into live orders that a broker can execute, turning your algorithm into an active market participant.

Phase 9: Execution Strategies

This phase revisits market microstructure with a focus on the strategic aspect of execution.

Contributing Modules & Lectures

- **Market Microstructure for Trading (MMT)**
 - The foundational lecture of this module is introduced in the '**Laying the Foundations**' phase. Here, the focus shifts to strategic execution.

AHA! MOMENT: Your Trades Move the Market

A crucial mental shift for any trader is realizing that you are not just a passive observer of prices.

When you trade, especially with size, your own orders affect the market by consuming liquidity. This "market impact" is a self-inflicted cost. The goal of sophisticated execution algorithms is to minimize this cost by trading intelligently.

- **Intra-Module Lecture Progression:**

- **MMT-01 and MMT-02 to MMT-03:** By introducing advanced "special" orders that provide the tools needed for more strategic execution, **MMT-03** builds on the knowledge of advanced orders from **MMT-02** by discussing how to apply them in the context of managing large trades to minimize market impact.
- **MMT-03 to MMT-04:** After learning the execution strategic concepts in **MMT-03**, **MMT-04** provides you with the quantitative models to formally measure and analyze whether your execution strategy was successful with respect to order execution knowledge.

 QUANT INSIGHT: Execution as a Source of Alpha

In the world of trading, minimizing costs is a form of alpha. This phase focuses on the strategy of *how* you trade, using advanced order types and quantitative models to reduce market impact and slippage, which can be the difference between a profitable and unprofitable system.

Phase 10: Go Live!

This is the transition from simulation to the real world, where your strategies interact with the live market.

Contributing Modules & Lectures

- **Trading & Backtesting Platforms (TBP)**

- The lectures in this module are explained in detail in the '**Create Scripts to Trade Algorithmically**' phase. This phase is the culmination of that module, focusing on **TBP-04 & 05**, where you use tools like IBridgePy to deploy, manage, and troubleshoot strategies in a live environment.

 QUANT INSIGHT: The Final Step

Do not be discouraged while starting to trade live and see no results. The transition from backtesting to live trading is one of the most challenging steps for any quant. This phase prepares you for that moment, focusing on the tools and procedures needed to deploy, manage, and troubleshoot your algorithms in a live, unpredictable market environment.

Phase 11: Performance Evaluation

This phase is actually integrated throughout the program, focusing on accurately measuring the performance of your strategy.

Contributing Lectures

- **EFS-01: Strategy Building in Equities:** Here you'll be introduced to foundational metrics like the Sharpe ratio and maximum drawdown.
- **OTS-05: Hedging and Trade Evaluation:** This lecture covers performance measurement specific to options, where you must properly account for the costs and benefits of hedging.

QUANT INSIGHT: Measuring What Matters

True performance is more than total profit. This program teaches you to use a sophisticated suite of risk-adjusted metrics to objectively evaluate your strategies, understand their true risk profile, and make informed decisions about their future. Because it's not only about strategy returns, but also about strategy volatility. The latter, with proper risk management, can give you a smooth equity curve.

Phase 12: Get a Quant Job

The entire curriculum is designed to build the comprehensive skill set required for a career in algorithmic trading. The capstone project is the final step, demonstrating your ability to integrate these skills.

Conclusion: The Capstone Project – Bringing It All Together

You can undertake complex projects that synthesize knowledge from across all 12 phases.

For example, building and deploying a **Reinforcement Learning (RL) trading agent** requires you to integrate skills from the entire program:

- You will master the **sourcing, engineering, and modeling of features** (Phase 2), utilizing knowledge from the **MLT** module.
- The agent's intelligence is derived from applying **neural network and adaptive learning concepts** (Phase 2), specifically from **MLT-03** and **MLT-07**.
- For project robustness and scalability, you will **structure your code and backtest it** (Phase 5), as taught in the **DMP** module.

- Deployment of your RL agent will be guided by **risk management principles** (Phase 3) from the **PRM** module.
- Finally, to **create scripts and go live** (Phases 8 & 10), you will utilize **API knowledge** from the **TBP** module to connect your agent to the market.

 QUANT INSIGHT: Demonstrating Your Expertise

The ultimate goal is to synthesize all 11 preceding phases into a single, comprehensive project. The capstone project is your opportunity to demonstrate professional-level competence by taking a complex strategy from ideation and backtesting all the way to a deployment-ready state, proving your readiness for a career in algorithmic trading.

Final Note

Please refer to this guide, the course map, and the accompanying document whenever you need orientation within the EPAT curriculum. If you have any questions, please contact us; we will be glad to assist you.

Ready?

Set?

Go Algo!